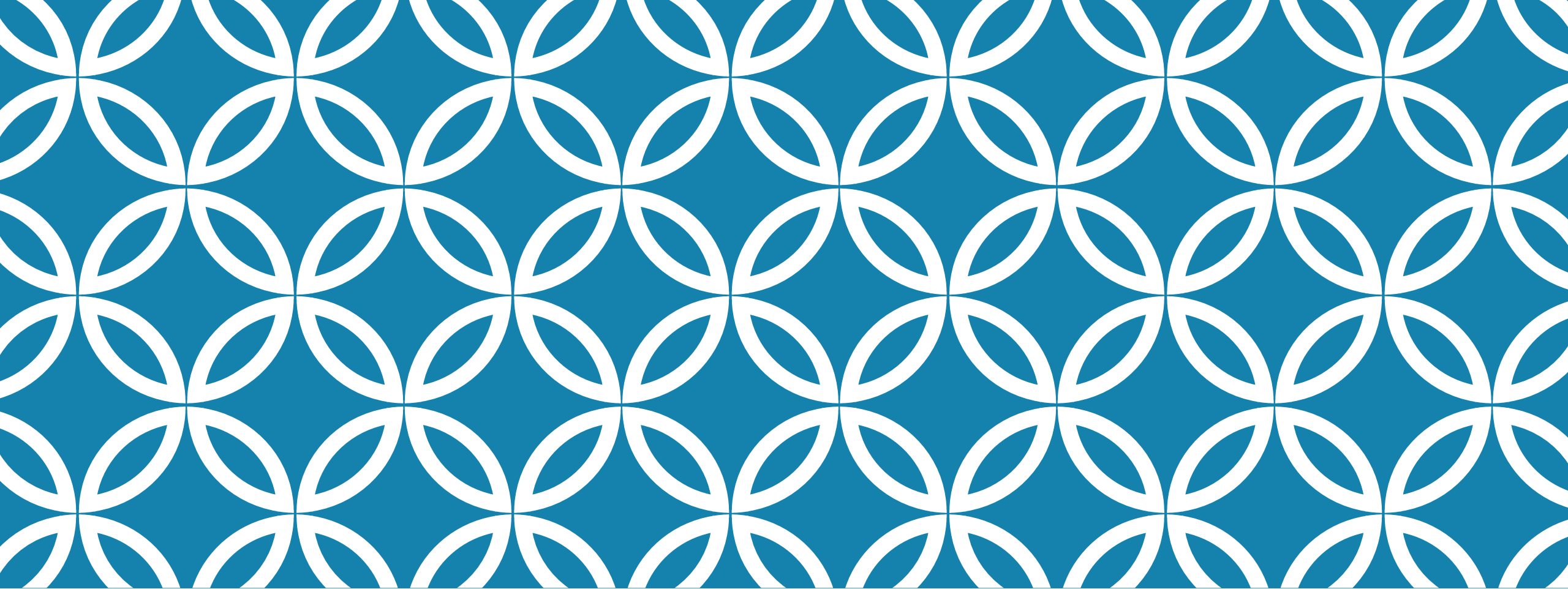




STATIC PROGRAM ANALYSIS

Lecture 4 – Refactoring problems
and AST limitations

ABSTRACT SYNTAX TREE

- Using visitor pattern can find more accurate answer to decision questions
- Knowing AST structure helps with answering question
 - Using a block visitor instead of visitors for each type of statement to count number of lines

SE REFACTORING PROBLEM

Renaming a given program variable

1. Problem statement
2. Rephrase it as a decision problem
3. Instantiate a SA
 - Method scope
 - Variable scope
 - Field of a class
4. Answer the decision problem

VARIABLE RENAMING

A 2-parts problem

1. Finding nodes to be renamed
2. Modifying the nodes

ECLIPSE AST MODIFIER/REWRITER

Rewriter is a special class `ASTRewriter` that takes in the ast to be modified.

```
CompilationUnit cu = (CompilationUnit) parser.createAST(null);  
ast = cu.getAST();  
rewriter = ASTRewrite.create(ast);
```

The same as for the AST
visitor

Three way to make changes:

- Modifying a node: `rewriter.replace(node, newNode, null);`
- Removing a node: `rewriter.remove(node, null)`
- Adding a node:
 - `ListRewrite listRewrite = rewriter.getListRewrite(node, CompilationUnit.IMPORTS_PROPERTY);`
`listRewrite.insertFirst(newNode, null);`

Changes depending on
the type of node

Let's look at `Rewriting.java`

ANALYSIS DESIGN

Let's sketch it together.

For a given method m_i , a given block where the variable is defined b_i , change variable's name from s_1 to s_2 . (Left as a HW 1 assignment)

SE CODE QUALITY PROBLEM

All variables must be initialized at the same time as they are declared.

- Instead of yellow warning, let's find those statements that are not initialized.
- Decision problem?
- Analysis design?

SE QUALITY PROBLEM

All assignments to variables must be used somewhere

- Somewhat challenging to do it over AST – need to remember way too much information
- Easier way is to propagate this information over a control flow graph CFG
- Reaching definition – a classic data-flow analysis
- WALA builds a CFG for you



NEXT TIME

WALA set-up

Obtaining CFG and iterating over it.

Reaching definitions analysis